

Onatola Timilehin Faruq

Blockchain Architect | Solutions Engineer | Protocol Designer

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Executive Summary

Blockchain architect with deep expertise across L1/L2 ecosystems, specializing in DeFi protocol design and cross-chain infrastructure. Production experience architecting solutions on Ethereum, Arbitrum, Base, Solana, Hedera, and Cosmos. UH17 graduate (Uniswap Hook Incubator, 15% acceptance rate) with direct Uniswap Labs collaboration on protocol-level tooling. Ships well-documented open-source projects with comprehensive technical specifications, API references, and implementation guides. Strong communicator with track record translating complex Web3 concepts into clear documentation and developer-focused content.

Core Competencies: Solution Architecture • Technical Documentation • DeFi Protocol Design • Cross-Chain Infrastructure • Security Review • Stakeholder Communication • Requirements Analysis

Technical Expertise

L1/L2 Ecosystems: Ethereum • Arbitrum • Base • Optimism • Solana • Hedera • Cosmos/Osmosis • Stacks • Mina

DeFi Protocols: Uniswap v3/v4 • Aave v3 • Curve v1/v2 • GMX • Chainlink (Oracles, VRF, Automation, CCIP)

Smart Contracts: Solidity (Advanced) • Rust/Anchor (Intermediate) • Noir • Clarity

Security: Contract Auditing • Slither • Fuzzing • Formal Verification • MEV Protection • Threat Modeling

Full-Stack: React • Next.js 14 • TypeScript • Node.js • PostgreSQL • Redis • GraphQL • WebSocket

Infrastructure: Foundry • Hardhat • IPFS/Arweave • Docker • GitHub Actions • CI/CD

Key Management: HD Wallets (BIP-39/32/44) • Multi-sig Patterns • Custodial/Non-custodial Architecture

Architecture & Documentation

Solution Design: Architected end-to-end DeFi systems integrating multiple protocols (Aave, Uniswap, Curve, GMX) with documented trade-offs, risk assessments, and implementation specifications. Experience evaluating on-chain vs. off-chain architecture decisions for financial protocols.

Technical Writing: Created 70KB+ documentation for OmniSwap SDK including API references, integration tutorials, architecture diagrams, and 5 working examples. All major projects include comprehensive README, CHANGELOG, and CONTRIBUTING guides. Won Mina Foundation Content Creation Track for developer-focused technical content.

Open Source: 11 public repositories on GitHub with full documentation and CI/CD pipelines. Active in community feedback loops, building in public with iteration based on developer input.

Security Documentation: Conducted security reviews on 15+ contracts, documenting vulnerabilities (reentrancy, front-running, oracle manipulation) with remediation recommendations. Experience producing threat models and risk assessments.

Professional Experience

Blockchain Engineer | DeFiConnectCredit | Remote | Jan 2025 – Present

- Architected modular smart contract systems integrating Aave v3, Uniswap v3, GMX perpetuals, and Curve pools, documenting architecture decisions and trade-offs
- Designed cross-chain asset management infrastructure reducing bridging latency by 40%
- Established testing framework with 95%+ coverage; identified and documented 3 critical vulnerabilities pre-deployment with risk assessments
- Produced technical specifications for gas optimization strategies achieving 35% cost reduction
- Built deployment automation suite with documentation, cutting deployment time from 2 hours to 15 minutes

Blockchain Development Engineer (Intern) | DeFiConnectCredit | Remote | Jun 2024 – Dec 2024

- Contributed to core lending protocol architecture: collateral management, liquidation mechanisms, interest rate models
- Conducted security reviews on 15+ contracts, producing vulnerability reports with severity ratings and remediation steps
- Designed and deployed ERC721 incentive system with Merkle tree proofs for 500+ users
- Developed comprehensive test suites with edge case documentation achieving 90%+ coverage

Featured Projects

OmniSwap SDK

Cross-Chain Swap SDK | TypeScript | github.com/Leihyn/omniswap-sdk

Architected production-grade SDK enabling atomic swaps across 6 heterogeneous blockchains.

- Designed HTLC-based swap protocol with documented security model and failure scenarios
- Built chain-specific adapters for Zcash, Osmosis, Mina, Aztec, Miden, Fhenix with unified interface specification
- Created 70KB+ documentation: architecture overview, API reference, integration tutorials, 5 working examples
- Established CI/CD pipeline with automated testing and release process

Sentiment – Uniswap v4 Dynamic Fee Hook

UHI7 Incubator Capstone | Solidity | github.com/Leihyn/Sentiment

Designed dynamic fee hook for Uniswap v4 implementing counter-cyclical fees based on market sentiment.

- Architected fee adjustment algorithm (0.25%-0.44%) with documented economic model and trade-offs
- Designed multi-keeper architecture using Chainlink Automation + Gelato for redundancy
- Collaborated directly with Uniswap Labs on protocol-level design decisions
- Graduated from UHI7 (15% acceptance rate) with production-ready implementation

Comic Pad – NFT Marketplace

Hedera Hashgraph | Production | github.com/Leihyn/comicPad

Solo-architected full-stack NFT publishing platform from requirements to production deployment.

- Designed HTS token system supporting 3 tokenization models: full-issue, paginated, collection series
- Architected WebSocket infrastructure handling 1K+ concurrent users with <50ms latency
- Designed hybrid IPFS/Arweave storage with 99.9% availability and automatic failover
- Built complete solution: smart contracts, React/Next.js frontend, real-time backend, wallet integrations

OmniShield Wallet

Multi-Chain HD Wallet | TypeScript | github.com/Leihyn/omnishield-wallet

Designed HD wallet architecture supporting 6 blockchains with secure key management.

- Architected BIP-39/32/44 key derivation system across heterogeneous chain types
- Designed chain-agnostic adapter interface for unified balance queries and transactions
- Built cross-platform solution: React web interface and React Native mobile app

TruthBounty – Reputation Protocol

BNB Chain | Solidity/Next.js | github.com/Leihyn/TruthBounty

2nd place at Seedify Prediction Market Hackathon. Designed decentralized reputation system for prediction markets.

- Architected accuracy tracking system across PancakeSwap & Polymarket predictions
- Designed soulbound NFT credential system for on-chain reputation
- Created TruthScore algorithm with documented scoring methodology

Cybria Bridge

Cross-Chain Bridge | EVM/Python | github.com/Leihyn/cybria-bridge

Designed cross-chain bridge architecture with multi-signature security model.

- Architected 3-of-5 multi-sig validator system with documented trust assumptions
- Designed message passing protocol with security considerations and failure modes

Additional Projects

Project	Tech	Description
USDCx-Lending	Clarity	Lending protocol architecture with interest rate models and liquidation mechanics
BankOfCronos	Solidity	AI-powered DeFi protocol design with automated hedging strategies
Nocturne	Noir/Solana	Private payments architecture using ZK proofs and stealth addresses
TerraCred	Hedera	Real estate DeFi design with property-backed collateralization
CelestiFi	TypeScript	Whale activity tracking system with sub-second alert architecture

Achievements & Certifications

2nd Place – Seedify Prediction Market Hackathon *TruthBounty: Designed decentralized reputation protocol for prediction markets*

Mina Foundation Content Creation Track Winner – Zypherpunk Hackathon *Technical content creation translating complex Web3 concepts for developer audiences*

Uniswap Hook Incubator (UHI7) – Graduate 15% acceptance rate • Protocol-level design • Direct Uniswap Labs collaboration

Cyfrin Updraft – DeFi Security & Protocol Engineering Aave v3 • Uniswap v3/v4 • GMX • Curve v1/v2 • Smart Contract Security

Hashgraph Developer Certification – The Hashgraph Association *HTS/HCS Architecture • Mirror Node Integration • Production Deployment*

Education

Bachelor of Science in Physiotherapy University of Ibadan, College of Medicine | Graduated 2024

Languages

Human: English (Native) • Spanish (Conversational)

Programming: Solidity • TypeScript • Rust • Clarity • Noir

Community & Open Source

- **11 public repositories** on GitHub with comprehensive documentation, CI/CD, and examples
 - Active in Discord, X, and GitHub; comfortable working in public with community feedback
 - All major projects include architecture docs, README, CHANGELOG, and CONTRIBUTING guides
 - Track record producing developer-focused content and technical specifications
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Portfolio: faruukku.vercel.app | github.com/Leihyn | References available upon request